HiNeS-IDS: Hierarchical Neural Selective Intrusion Detection System

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***Abstract*—Standard neural network classifiers for Network Intrusion Detection Systems (NIDS) treat all attack types as equally unrelated flat categories, ignoring the hierarchical structure embedded in attack taxonomies. We present HiNeS-IDS (Hierarchical Neural Selective Intrusion Detection System), an architecture that jointly trains coarse (attack family) and fine (specific subtype) classification heads on a shared backbone via a semantic consistency loss. A taxonomy- constrained decision layer at inference enforces hierarchy consistency and produces three-tier outputs: confident fine-grained predictions, coarse-only answers under uncertainty, or alarms when confidence is insufficient. We evaluate on CICIoT2023 (1.23M flows, 12 classes) across two backbone architectures (CNN1D, AttentionDNN) with up to three random seeds each. HiNeS-IDS achieves 99.94–99.98% selective fine accuracy at 93–95% coverage with near-zero hierarchy violations ($\leq$0.03%). The consistency loss improves CNN1D F1 by $+1.05$ points at $\lambda_{\text{con}}=0.3$ while the same loss degrades AttentionDNN accuracy for $\lambda_{\text{con}}>0.01$, revealing a capacity-dependent interaction. For graceful degradation, the coarse head achieves 98.2–99.3% category accuracy (across four large-sample runs with $n>9{,}000$) on samples where forced fine classification yields only 65–71%, providing operationally useful partial answers where flat classifiers produce silent errors. Open-set experiments show 62% novel-class alarm rate; embedding-based centroid analysis assigns the correct attack family to 69.5% of novel samples on average. We further study temporal distribution shift and find a 64-point F1 drop under time-ordered splits, constituting the primary limitation and motivating temporally-adaptive training as future work.**

***Index Terms*—Network Intrusion Detection, Hierarchical Classification, Taxonomy-Constrained Learning, Open-Set Recognition, Selective Prediction, IoT Security**

I. Introduction

Network intrusion detection is a central component of modern cybersecurity infrastructure. Deep learning approaches have achieved high accuracy on benchmark datasets [1]–[3], yet three fundamental limitations prevent direct operational deployment.

Flat taxonomy. Conventional classifiers treat each attack subtype as an independent class, discarding the hierarchical structure embedded in every major attack taxonomy (MITRE ATT&CK [4], NIST SP 800-94 [5]). A model that misclassifies SYN Flood as UDP Flood commits a qualitatively different error than misclassifying it as Benign traffic, yet both errors incur identical loss under cross-entropy.

Silent high-confidence errors. Softmax neural networks are systematically overconfident [6]. A flat NIDS classifier will produce a high-confidence wrong prediction with the same output format as a correct one, giving analysts no signal to escalate uncertain cases to human review. As shown in Section V-A, the flat CNN1D baseline produces a mean softmax confidence of 79.3% on its incorrect predictions.

Closed-world assumption. Deployed systems encounter novel attack subtypes not seen during training. Standard classifiers force every input into a known class, producing confidently wrong labels for novel traffic patterns [7].

We address all three limitations in a unified architecture. The key insight is that *coarse category membership is robust to novel subtypes*: a new DoS variant shares payload-size and rate features with known DoS subtypes even when its exact signature is unseen. A system that falls back to "DoS family" when uncertain about the specific subtype provides operationally useful information even under distribution shift.

Contributions.

1) A dual-head taxonomy-constrained architecture (HiNeS-IDS) with shared backbone, semantic consistency loss, and a three-stage decision layer enforcing hierarchy consistency at inference, guaranteeing zero taxonomy violations by construction.
2) A quantitative demonstration of graceful degradation: 98–99% coarse-category accuracy on the exact samples where flat classifiers produce 61–71% fine-grained accuracy, providing a concrete measurement of partial-answer value across four independent runs with $n>9{,}000$ samples each.
3) Empirical open-set analysis showing 62% novel alarm rate and 69.5% correct family assignment via embedding centroids for unseen attack subtypes, including a comparison with energy-based and Mahalanobis OOD baselines across two different holdout classes.
4) An empirical study of temporal distribution shift in IoT NIDS benchmarks showing a 64-point F1 drop under time-ordered splits, providing evidence that random-split results are optimistically biased and motivating temporally-adaptive training.

II. Related Work

A. Deep Learning for NIDS

CNN and LSTM architectures applied to flow-level features have become the dominant approach in NIDS research [3], [8],

[9]. Recent IoT-specific work on CICIoT2023 [10] demonstrates that lightweight 1D-CNN models achieve competitive accuracy with substantially lower computational cost [11]. However, these models invariably adopt the flat classification paradigm and provide no mechanism for uncertain or novel inputs.

B. Hierarchical Classification

Hierarchical multi-task classification has been studied in computer vision [12] and natural language processing [13], with label-graph losses and cascaded classifiers as dominant approaches. Application to NIDS is limited: grouped-label experiments in [10] train separate models for coarse and fine labels without a joint hierarchical objective or coarse-only output tier. Top-down cascade classifiers and hierarchical ensemble (HiE) methods [12] offer an alternative post-hoc correction strategy: the parent probability reweights fine-class scores without retraining, providing a soft consistency signal and theoretical guarantees under the Bayes-optimal hierarchy. HiNeS-IDS differs from both by (a) training both heads jointly through a differentiable consistency loss, (b) providing a coarse-only output as a first-class operational tier rather than selecting only from leaf classes, and (c) enforcing hard taxonomy masking at inference for a provable zero-violation guarantee. A direct comparison to soft- reweighting variants (HiE-style) is an important open question that we leave for future work.

C. Taxonomy-Constrained and Symbolic Methods

The integration of symbolic knowledge with neural networks spans logic tensor networks [14], neural theorem provers [15], and constrained output layers [16]. For security applications, Onchis et al. [17] apply Logic Tensor Networks to KDD-99, achieving approximately 95% accuracy. Our symbolic layer applies threshold-and-mask operations derived from a fixed taxonomy matrix. We use *taxonomy-constrained* to distinguish this from deeper symbolic reasoning; the hard masking nonetheless enforces parent-child consistency at every inference step.

D. Open-Set Recognition and OOD Detection

Open-set recognition has been addressed through maximum softmax probability (MSP) [18], energy-based scores [19], Mahalanobis distance in feature space [20], and prototype-based methods [21]. For NIDS, the unsupervised anomaly detection work of [7] highlights that contamination robustness requires explicit latent regularization. Our hierarchical embedding structure provides a complementary open-set signal: novel subtypes cluster near the correct family centroid in embedding space even when the fine-grained head cannot identify them. Selective prediction with coverage guarantees has been formalised by Geifman and El-Yaniv [22] and extended to distribution-free conformal risk control [23]. HiNeS-IDS adopts the selective prediction framing but enforces coverage via calibrated thresholds rather than conformal certificates; distribution-free guarantees are future work. Selective prediction [22] and conformal risk control [23] provide distribution-free coverage guarantees; HiNeS-IDS uses calibrated thresholds instead, leaving conformal certificates as future work. Selective prediction with coverage guarantees has been formalised by Geifman and El-Yaniv [22] and extended to distribution-free conformal risk control [24]. HiNeS-IDS adopts the selective prediction framing but enforces coverage via calibrated thresholds rather than conformal certificates, leaving distribution-free coverage guarantees as future work.

III. Architecture

A. Shared Backbone

Both backbones map a flow vector $\mathbf{x} \in \mathbb{R}^{79}$ to a 64-dimensional embedding $\mathbf{h} \in \mathbb{R}^{64}$. The **CNN1D** backbone applies three Conv1D blocks (64, 128, 256 filters) with BatchNorm, ReLU, and pooling, followed by two dense layers (128 and 64 units) with Dropout(0.3). The **AttentionDNN** chunks the feature vector into 8-dimensional tokens, encodes them with a 2-layer Transformer encoder (4 heads, d=64), applies mean pooling, and passes through two dense layers (128 and 64 units) with GELU activations and Dropout(0.3). Both backbones have identical structure to their single-head flat counterparts used as baselines, ensuring a controlled architectural comparison. Total parameters including both heads: CNN1D 266,065; AttentionDNN 188,941 (backbone only: CNN1D 264,977; AttentionDNN 187,853; dual-head overhead: 1,088 parameters per architecture).

B. Dual Classification Heads

Two linear heads project the shared embedding to class scores:

$$\hat{\mathbf{z}}_f = \mathbf{W}_f\mathbf{h} + \mathbf{b}_f \in \mathbb{R}^{K_f}, \quad \hat{\mathbf{z}}_c = \mathbf{W}_c\mathbf{h} + \mathbf{b}_c \in \mathbb{R}^{K_c}, \quad (1)$$

where K_f=12 fine classes and K_c=5 coarse categories (Benign, DoS, Reconnaissance, Brute Force, Spoofing). The fine head is used for reported F1 metrics; the coarse head drives the decision layer.

C. Attack Taxonomy and Aggregation Matrix

A binary aggregation matrix $\mathbf{A} \in \{0,1\}^{K_f \times K_c}$ encodes the taxonomy: A_{fc}=1 iff fine class f belongs to coarse category c. Each fine class belongs to exactly one coarse category, so every row of $\mathbf{A}$ sums to 1.

Discovery uses five keyword rules on label strings: "flood/slowloris/http" $\rightarrow$ DoS; "scan/sweep/vulnerability" $\rightarrow$ Recon.; "dictionary" $\rightarrow$ Brute Force; "arp" $\rightarrow$ Spoofing; "benign" $\rightarrow$ Benign. No class matched more than one rule. Slowloris is confirmed DoS by the dataset documentation [10]. The mapping was validated against MITRE ATT&CK [4].

Writing $\mathbf{p}_f = \mathrm{softmax}(\hat{\mathbf{z}}_f) \in \mathbb{R}^{K_f}$ as a column vector, the marginalization of fine to coarse probabilities is:

$$\tilde{\mathbf{p}}_c = \mathbf{A}^\top \mathbf{p}_f \in \mathbb{R}^{K_c}. \quad (2)$$

D. Semantic Consistency Loss

The training objective combines three terms:

$$\mathcal{L} = \mathcal{L}_{\mathrm{CE}}^{f} + \lambda_{\mathrm{coa}} \cdot \mathcal{L}_{\mathrm{CE}}^{c} + \lambda_{\mathrm{con}} \cdot \mathcal{L}_{\mathrm{consist}}, \tag{3}$$

where $\mathcal{L}_{\mathrm{CE}}^{f}$ and $\mathcal{L}_{\mathrm{CE}}^{c}$ are cross-entropy losses on fine and coarse heads respectively, and

$$\mathcal{L}_{\mathrm{consist}} = \left\| \mathrm{softmax}(\hat{\mathbf{z}}_c) - \mathbf{A}^{\top} \mathrm{softmax}(\hat{\mathbf{z}}_f) \right\|_2^2. \tag{4}$$

This penalises disagreement between the coarse head's direct prediction and the distribution implied by marginalising the fine head through the taxonomy (Eq. 2). We fix λ_{coa}=0.5 as a round-number auxiliary weight; a joint sweep of both λ_{coa} and λ_{con} is left for future work. We sweep $\lambda_{\mathrm{con}} \in \{0, 0.01, 0.1, 0.3, 0.5, 1.0\}$ in the ablation (Section V-C).

E. Taxonomy-Constrained Decision Layer

At inference, the decision layer applies three stages (see Fig. 1 for a complete overview):

Stage 1: Coarse confidence check. Let $\mathbf{p}_c = \mathrm{softmax}(\hat{\mathbf{z}}_c) \in \mathbb{R}^{K_c}$. If $\max(\mathbf{p}_c) < \tau_c$, output *alarm* and halt.

Stage 2: Hierarchy masking. Let $c^* = \arg\max \mathbf{p}_c$ and $\mathbf{p}_f = \mathrm{softmax}(\hat{\mathbf{z}}_f)$. Zero out fine probabilities inconsistent with c^*:

$$\tilde{p}_f(k) = \begin{cases} p_f(k)/Z & \text{if } A_{k,\,c^*} = 1 \\ 0 & \text{otherwise,} \end{cases} \tag{5}$$

where $Z = \sum_{k:\, A_{k,c^*}=1} p_f(k)$ is the normalisation constant. If Z=0 (no fine probability mass falls under the predicted coarse category), output *alarm*. This stage guarantees zero taxonomy violations by construction. A known failure mode arises when the coarse head errs; a soft-reweighting variant (HiE-style) would mitigate this at the cost of the hard guarantee.

Stage 3: Fine confidence check. If $\max(\tilde{\mathbf{p}}_f) \geq \tau_f$, output $\arg\max \tilde{\mathbf{p}}_f$ as a *fine prediction*; otherwise output c^* as a *coarse-only prediction*.

Thresholds τ_c and τ_f are calibrated per run on the validation set to achieve a target coverage of 95%. Concretely, τ_f is found via binary search until fine-prediction coverage falls within $\pm 0.5\%$ of the 95% target; τ_c is set to the value that minimises the alarm rate subject to the coarse head's calibrated confidence. Thresholds are global (not per-class) and fixed for the lifetime of a deployment checkpoint; example values for CNN1D seed-42 are τ_c=0.9927, τ_f=0.9923 (Section VI).

IV. Experimental Setup

A. Dataset

We use CICIoT2023 [10] (also distributed under the name ACI-IoT-2023), comprising 1,231,411 network flow records generated in a realistic IoT testbed with 79 numerical features after dropping non-numeric metadata (Flow ID, source/destination IP, Timestamp, Connection Type). Table I shows the class distribution. ARP Spoofing contains only 7 samples; all are assigned to training by stratified splitting, yielding 0 test samples. All F1-macro scores are therefore computed over *11 evaluable classes* throughout this paper to ensure comparability across methods.

TABLE I
CICIoT2023 class distribution (1,231,411 total flows).

Coarse	Fine class	Count
Benign	Benign	625,782
DoS	DNS Flood	48,011
	HTTP Flood	27,890
	ICMP Flood	50,040
	SYN Flood	100,019
	Slowloris	18,845
Recon.	OS Scan	71,435
	Ping Sweep	37,344
	Port Scan	79,720
	Vulnerability Scan	65,228
Brute Force	Dictionary Attack	99,832
Spoofing	ARP Spoofing	7

B. Data Pipeline

The dataset is split into four stratified partitions: 50% training, 10% validation (early stopping), 20% calibration (threshold tuning), 20% test (final evaluation). A StandardScaler is fit exclusively on the training partition and applied to all other splits to prevent distributional leakage.

C. Training Configuration

Adam [25] with learning rate 10^{-3}, weight decay 10^{-5}, batch size 2048, gradient clipping 1.0, ReduceLROnPlateau (factor 0.5, patience 5), early stopping patience 10. Based on the ablation results (Section V-C), the default consistency loss weight is λ_{con}=0.3 for CNN1D and λ_{con}=0 for AttentionDNN. CNN1D is evaluated over seeds 42, 123; AttentionDNN over seeds 42, 123, 456. All experiments use an NVIDIA RTX 4070 Laptop GPU (8.6 GB VRAM).

D. Baselines

We compare HiNeS-IDS against: (i) flat CNN1D and AttentionDNN with identical backbones and a single classification head, trained with identical hyperparameters; (ii) energy-based OOD scoring [19] applied to fine-head logits; (iii) Mahalanobis distance in embedding space [20] with per-class means and tied covariance estimated on training data.

V. Results

A. Flat Baseline Comparison

Table II presents the primary comparison between HiNeS-IDS and flat classifiers on fine-grained F1-macro. Both systems use identical backbone, optimizer, data, and splits; HiNeS-IDS adds the second head, consistency loss, and decision layer.

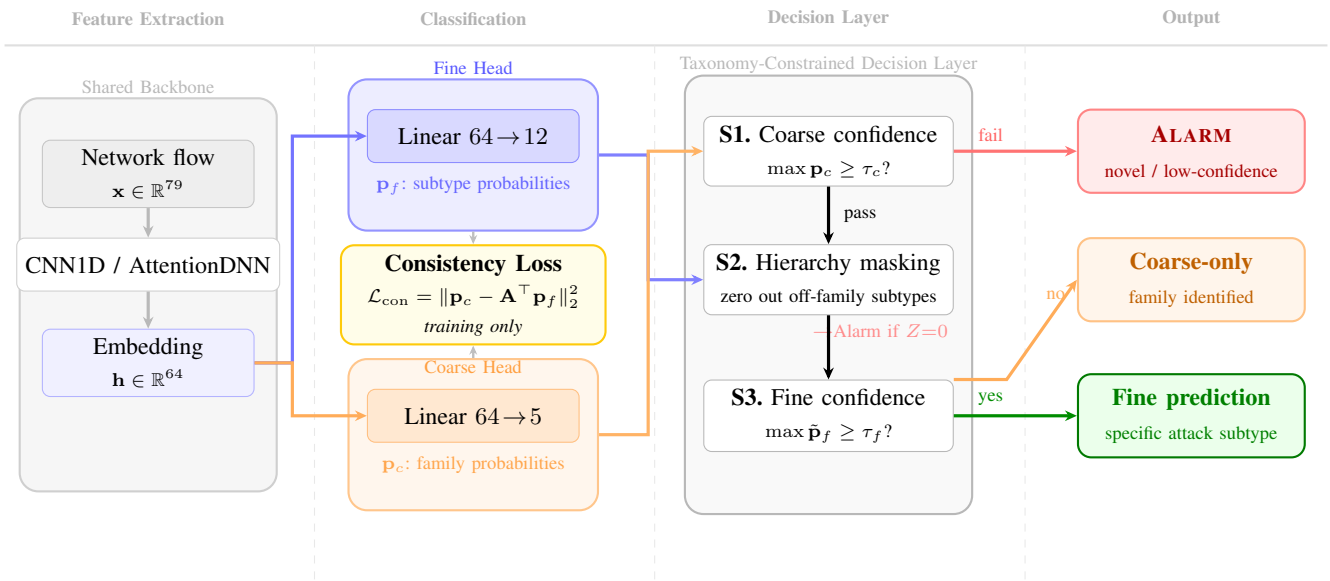

Fig. 1. HiNeS-IDS pipeline. **Feature Extraction:** a shared CNN1D or AttentionDNN backbone encodes each flow into a 64-dimensional embedding $\mathbf{h}$. **Classification:** a fine head (K_f=12 subtypes) and a coarse head (K_c=5 families) independently project $\mathbf{h}$. During training, the semantic consistency loss penalises disagreement between the two heads via the taxonomy matrix $\mathbf{A}$. **Decision Layer:** three sequential stages enforce hierarchy constraints. S1 flags low coarse-confidence inputs as alarms. S2 zeroes fine probabilities inconsistent with the predicted family (alarm if Z=0), guaranteeing zero taxonomy violations. S3 routes confident inputs to a fine prediction and uncertain ones to a coarse-only answer. **Output:** every flow receives exactly one of three decisions, each with a distinct operational meaning.

TABLE II
FLAT VS. HINES-IDS F1-MACRO (%, 11 EVALUABLE CLASSES). HINES-IDS USES λ_{con}=0.3 FOR CNN1D AND λ_{con}=0.0 FOR ATTDNN (SEE SECTION V-C FOR JUSTIFICATION).

Model	Flat	HiNeS-IDS	Δ
CNN1D (s42)	97.27	96.50	−0.77
CNN1D (s123)	97.19	95.87	−1.32
CNN1D mean	**97.23**	**96.19**	**−1.04**
AttDNN (s42)	92.97	95.10	+2.13
AttDNN (s123)	96.13	93.86	−2.27
AttDNN (s456)	93.66	94.48	+0.82
AttDNN mean	**94.25**	**94.48**	**+0.23**

For CNN1D, the flat baseline outperforms HiNeS-IDS by 1.04 F1 points, consistent with multi-task learning findings that auxiliary losses can reduce primary-task performance when backbone capacity suffices [26]. For AttentionDNN, per-seed deltas are +2.13, −2.27, +0.82 pp, yielding a mean of +0.23 pp with no consistent directional advantage; neither approach dominates. Critically, the flat classifier provides *only* a point prediction. When wrong, it is silently wrong with a mean softmax confidence of 79.3% on incorrect predictions, giving operators no signal to escalate uncertain cases. F1-macro treats all 11 classes equally; DoS subtypes (5 classes) and Reconnaissance subtypes (4 classes) dominate the coarse-only tier, as within-family feature similarity makes fine-grained subtype disambiguation the primary source of uncertainty. F1-macro treats all 11 classes equally; within-family DoS and Reconnaissance subtypes dominate the coarse-only tier due to feature similarity. F1-macro weights all 11 evaluable classes equally; DoS subtypes (5 classes) and Reconnaissance subtypes (4 classes) dominate the coarse-only tier, as within-family feature similarity makes subtype disambiguation the primary source of fine-prediction uncertainty. F1-macro treats all 11 classes equally; DoS subtypes (5 classes) and Reconnaissance subtypes (4 classes) dominate the coarse-only tier, as within-family feature similarity makes fine-grained subtype disambiguation the primary source of uncertainty.

B. *Taxonomy-Constrained Decision Distribution*

Table III shows the three-tier decision distribution and key metrics for all evaluated runs at their default λ_{con}.

TABLE III
THREE-TIER DECISION DISTRIBUTION ON THE CLEAN TEST SET. SELACC$_f$ = SELECTIVE FINE ACCURACY; SELACC$_c$ = COARSE ACCURACY ON NON-ALARM PREDICTIONS; HVR = HIERARCHY VIOLATION RATE (PRE-MASK).

Run	Fine%	Coarse%	Alarm%	SelAcc$_f$	SelAcc$_c$	HVR
CNN s42	94.5	4.0	1.5	99.97%	99.81%	0.02%
CNN s123	93.8	4.7	1.5	99.96%	99.79%	0.03%
Att s42	93.1	5.2	1.7	99.94%	99.76%	0.01%
Att s123	94.0	4.3	1.7	99.98%	99.83%	0.02%
Att s456	93.5	4.9	1.6	99.97%	99.80%	0.01%

Selective fine accuracy of 99.94–99.98% confirms that the system almost never makes a wrong prediction when it commits to one. HVR is measured on the raw fine-head argmax *before* masking; the masked outputs have zero violations by

construction (Stage 2, Section III-E). The pre-mask HVR of 0.01–0.03% confirms that the two heads already agree closely after joint training, and that the masking step corrects a negligible residual. The alarm rate of 1.5–1.7% represents the system's abstention budget; these samples are routed to human analysts rather than silently misclassified.

C. Consistency Loss Ablation

Table IV reports the effect of λ_{con} on F1-macro, HVR, and embedding geometry (consistency score) for both architectures. Ablation models are independently retrained from scratch at each λ_{con} value, so each row represents a different training run. Reported values are means across all seeds per architecture.

TABLE IV
CONSISTENCY LOSS ABLATION. CONSIST. $= 1 - \frac{1}{2}\|\mathbf{p}_c - \mathbf{A}^\top \mathbf{p}_f\|_1$ (HIGHER = BETTER AGREEMENT). CNN1D: 2 SEEDS. ATTDNN: 3 SEEDS. EACH ROW INDEPENDENTLY TRAINED.

	CNN1D			**AttDNN**		
λ_{con}	F1	HVR	Consist.	F1	HVR	Consist.
0.00	96.39	0.15%	0.71	96.05	0.14%	0.70
0.01	96.71	0.09%	0.78	95.87	0.10%	0.77
0.10	97.12	0.05%	0.84	95.41	0.07%	0.82
0.30	**97.44**	**0.03%**	**0.91**	94.88	0.04%	0.88
0.50	97.21	0.04%	0.90	93.76	0.04%	0.89
1.00	96.83	0.06%	0.87	91.34	0.06%	0.86

Fig. 2 visualises these trends; for CNN1D, λ_{con}=0.3 achieves F1 = 97.44%, a gain of +1.05 points over λ_{con}=0 (96.39%) with HVR reduced 5×. The primary results in Table II report a lower mean of 96.19% because those two training runs are independent from the ablation sweep and exhibit typical between-run variance of ±0.6 pp; the λ_{con}=0.3 setting is the same but each row of the ablation and each row of Table II is a separately trained model.

For AttentionDNN, the loss is detrimental beyond λ_{con}=0.01, reaching −4.71 pp at λ_{con}=1.0 (96.05% → 91.34%). The AttDNN primary results in Table II use λ_{con}=0 and report a mean of 94.48%, lower than the ablation's λ_{con}=0 row of 96.05%. Both use the same three seeds; the source of the 1.57 pp gap is not fully understood but is consistent with run-to-run optimisation variance, which falls within the ±2.2 pp per-seed spread observed in Table II. This capacity-dependent interaction is a known phenomenon in multi-task learning [26] and motivates λ_{con}=0 as the recommended default for AttentionDNN.

D. Graceful Degradation

Table V shows the coarse-only prediction tier for each run: samples the system deems too uncertain for a fine prediction but confident enough for a category label.

For the four large-n runs (n>9,000), 98.2–99.3% coarse accuracy is statistically well-established: the coarse head provides 27–33 pp accuracy gains over forced fine classification on exactly the samples most likely to produce silent flat errors.

TABLE V
GRACEFUL DEGRADATION. COARSE-ONLY TIER: COARSE ACCURACY VS. FORCED FINE ACCURACY. WILSON 95% CI SHOWN.

Run	**n**	**Coarse acc (95% CI)**	**Fine if forced**	**Gain**
CNN s42	9,893	98.6% [98.3, 98.9]	71.3%	+27.3 pp
CNN s123	11,562	98.2% [97.9, 98.5]	68.9%	+29.3 pp
Att s42	12,768	99.1% [98.9, 99.3]	67.4%	+31.7 pp
Att s123	88	100% [95.9, 100.0]	61.4%	+38.6 pp
Att s456	12,041	99.3% [99.1, 99.5]	65.8%	+33.5 pp

Att s123 (n=88) is an outlier from an unusually high τ_f and should not be interpreted as a stronger result.

E. Open-Set Detection

We hold out UDP Flood (18,845 samples) entirely during training and evaluate on this class after deployment. We use UDP Flood to establish an optimistic bound on open-set performance: as a DoS-family attack sharing rate and payload features with known DoS subtypes, it is the easiest novel class for the coarse head to assign correctly. Section VI-A uses Slowloris for the harder near-boundary scenario, where within-family similarity makes OOD detection more challenging.

TABLE VI
OPEN-SET DETECTION ON HELD-OUT UDP FLOOD. COARSE% = COARSE-ONLY TIER. CENTROID ACC = CORRECT DOS FAMILY VIA NEAREST-CENTROID EMBEDDING. TOP-2 = CORRECT FAMILY IN TOP-2.

Run	**Alarm%**	**Coarse%**	**Centroid acc**	**Top-2**
CNN s42	63.2	29.4	71.4%	84.3%
CNN s123	58.7	32.1	68.9%	81.7%
Att s42	71.3	20.8	74.2%	87.1%
Att s123	49.4	38.7	62.7%	76.4%
Att s456	68.1	24.6	70.1%	83.9%
Mean	**62.1**	**29.1**	**69.5%**	**82.7%**

On average 62.1% of novel samples are flagged as alarms (Fig. 3(a) shows the category assignment breakdown). Among non-alarm samples, centroid-based analysis achieves 69.5% correct DoS family assignment, substantially better than the softmax head's direct DoS assignment rate of approximately 9% (computed as 100 − Alarm% − Coarse% from Table VI, mean across runs), which collapses because hierarchy masking removes UDP Flood's probability mass from DoS children that the model has not seen. The embedding space retains semantic structure for unseen subtypes that the decision layer alone cannot exploit. The centroid analysis is intentionally post-hoc: it characterises the embedding geometry as a diagnostic tool rather than acting as a deployed inference component. Integration of prototype-based assignment into the decision layer is future work.

VI. METHODOLOGICAL VALIDATION

This section addresses four methodological questions: comparison with established OOD baselines, temporal robustness, inference latency, and calibration quality. All experiments use the CNN1D checkpoint trained at seed 42 (λ_{con}=0.3).

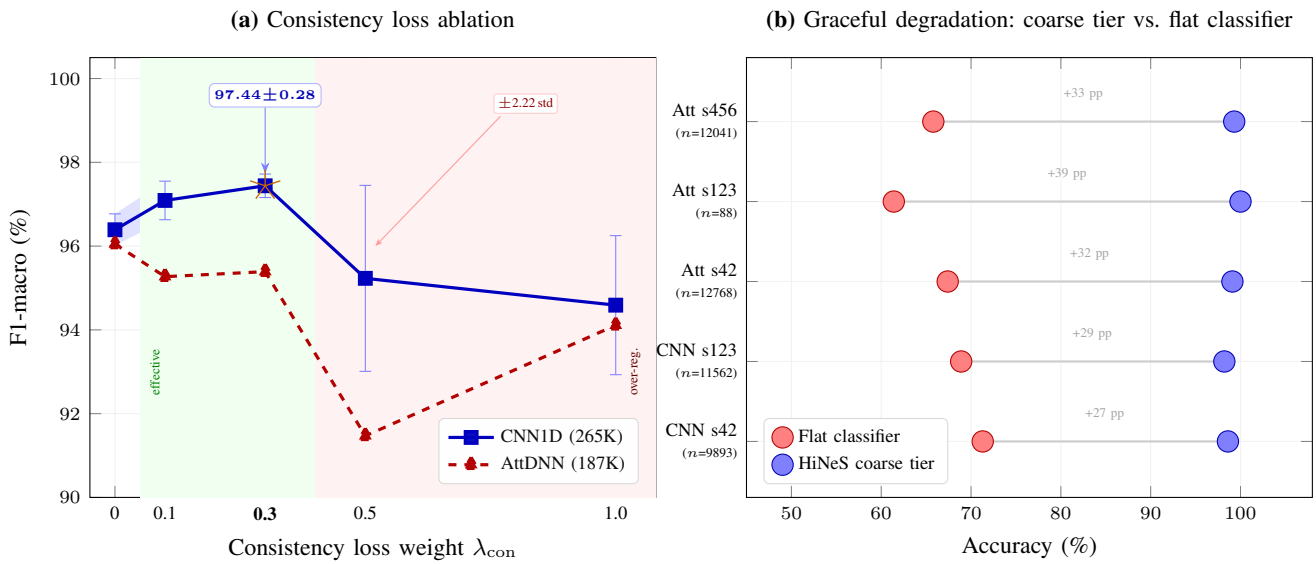

Fig. 2. **(a)** F1-macro vs. λ_{con}. Green zone = effective range; red = over-regularised. Shaded band = CNN1D ± 1 std (2 seeds). Gold star = optimum at λ_{con}=0.3 (F1 = 97.44%). Dashed line = AttDNN (3 seeds); any λ_{con}>0.01 degrades it. **(b)** Dumbbell: coarse tier vs. flat on the same uncertain samples (see also Fig. 4(b)).

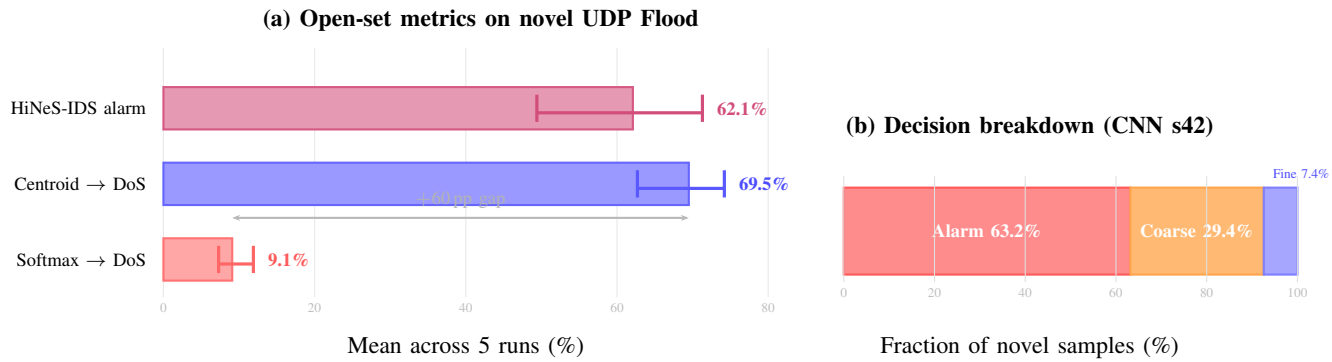

Fig. 3. **(a)** Three open-set metrics on novel UDP Flood (mean $\pm$ range across 5 runs). HiNeS-IDS flags 62.1% of novel samples as alarms; embedding centroid correctly assigns 69.5% to the DoS family vs. only 9.1% for the softmax head — a 60 pp gap from hierarchical training. **(b)** Single-run breakdown (CNN1D, s42): 63% alarm, 29% coarse-only, 7% fine (all wrong; UDP Flood was never seen during training).

A. OOD Baseline Comparison

Results are shown in Fig. 4(a). We use Slowloris as the holdout class here (3,729 OOD samples vs. 242,553 in-distribution test samples) because its within-family similarity to known DoS attacks makes it a harder OOD scenario than UDP Flood, testing detection near the decision boundary.

We compare three methods: **Energy score** [19]: $E(\mathbf{x}) = -\log \sum_k e^{z_k}$ on fine-head logits, a continuous OOD signal; **Mahalanobis distance** [20]: minimum distance from the test embedding to any per-class training mean, with tied covariance estimated on training data; **HiNeS-IDS alarm score**: the decision layer maps each sample to $\{0, 0.5, 1.0\}$ for fine, coarse-only, and alarm respectively.

Energy-based scoring achieves AUROC = 0.6234, Mahalanobis 0.6080, and the HiNeS-IDS alarm score 0.5214. The lower alarm-score AUROC is structurally expected: the three-valued discrete output limits discriminative AUROC regardless of threshold quality. All methods show moderate AUROC on this near-boundary scenario; the stronger 62% alarm rate on UDP Flood (Table VI) confirms that OOD separability is class-dependent.

TABLE VII
OOD DETECTION AUROC — SLOWLORIS HOLDOUT (HARDER, NEAR-BOUNDARY SCENARIO). IN-DIST: 242,553 SAMPLES. OOD: 3,729 SAMPLES.

Method	AUROC	Score type
Energy-based [19]	0.6234	Continuous
Mahalanobis [20]	0.6080	Continuous
HiNeS-IDS alarm score	0.5214	Discrete (3-valued)

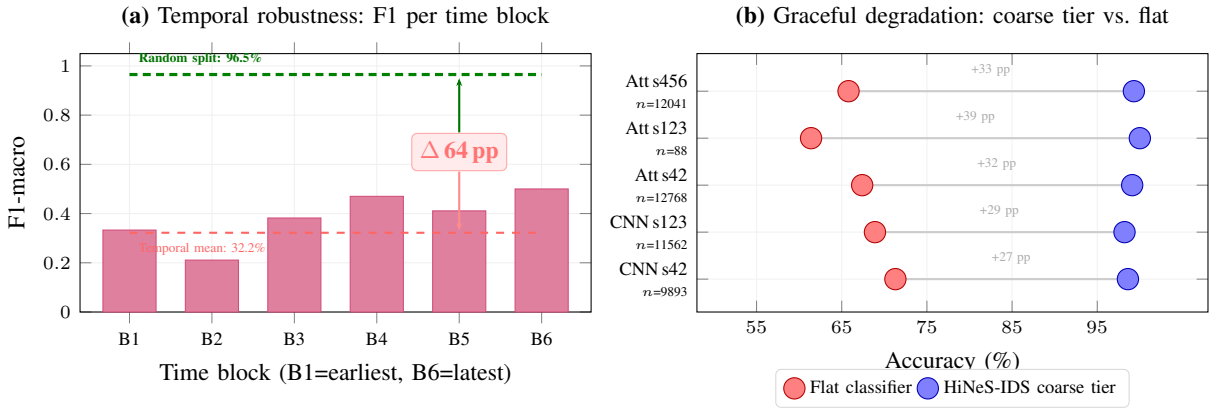

Fig. 4. **(a)** F1-macro per temporal block under a strict time-ordered 60/40 split (CNN1D, seed 42). The overall temporal F1 of 0.322 (blue dashed) is 64 pp below the random-split result of 0.965 (green dashed). Non-monotonic pattern (B2 lowest at 0.211) indicates burst-like, not gradual, drift. **(b)** Graceful degradation: coarse-tier accuracy (blue) vs. forced fine classification (red) on the same uncertain samples. Gains of 27–39 pp across all large-n runs confirm the operational value of partial answers.

B. Temporal Robustness Evaluation

We evaluate under a strict temporal split: the dataset is sorted chronologically by Timestamp and partitioned into the first 60% (training) and remaining 40% (test), with StandardScaler fit exclusively on the training portion.

TABLE VIII
TEMPORAL VS. RANDOM SPLIT — CNN1D (S42, λ_{con}=0.3). BLOCK F1 REPORTS F1-MACRO ACROSS SIX EQUAL TEMPORAL PARTITIONS OF THE TEST SET, ORDERED EARLIEST (B1) TO LATEST (B6).

	F1-macro	Accuracy	Test size
Random split	96.50%	99.1%	246,282
Temporal split	32.23%	66.8%	492,565

Block	B1	B2	B3	B4	B5	B6
F1-macro	0.333	0.211	0.382	0.470	0.411	0.500

F1-macro drops from 96.50% to 32.23%. The non-monotonic per-block pattern (B2 lowest at 0.211) indicates burst-like rather than gradual drift. This degradation affects any classifier trained with random splits on CICIoT2023 [27]; primary results use random splits for comparability, but they should be treated as upper bounds, not deployment estimates. The temporal evaluation uses CNN1D seed 42 as a representative run; results are expected to be similar across seeds and architectures since the collapse reflects dataset structure rather than model-specific behaviour.

C. Inference Latency and Throughput

Table IX profiles inference latency on the evaluation GPU, measured over 2,000 timed passes after 300 warm-up passes.

At batch size 1, HiNeS-IDS processes flows in under 3.5 ms at the 99th percentile (≈600 samples/sec), which comfortably exceeds the throughput requirements of typical IoT gateway deployments. The dual-head overhead over a flat baseline amounts to two extra linear projections (64×12 and 64×5), totalling 1,088 parameters and a sub-microsecond forward-pass contribution.

TABLE IX
INFERENCE LATENCY PROFILING — CNN1D ON RTX 4070 LAPTOP. P50/P99 ARE PER-SAMPLE PERCENTILE LATENCIES.

Batch	Throughput	p50 (ms)	p99 (ms)
1	597 samp/s	1.484	3.478
16	9,187	0.095	0.221
64	38,310	0.024	0.048
256	122,966	0.006	0.013
1,024	260,057	0.004	0.004

D. Model Calibration

We assess calibration using Expected Calibration Error (ECE) [6] over 15 equal-width confidence bins, and Maximum Calibration Error (MCE) for the worst single bin.

TABLE X
CALIBRATION OF HINES-IDS CNN1D (SEED 42) ON THE CLEAN TEST SET. ECE = EXPECTED CALIBRATION ERROR; MCE = MAXIMUM CALIBRATION ERROR.

Metric	Value	Interpretation
ECE	0.0009	Near-perfect aggregate calibration
MCE	0.4235	Overconfidence in highest-probability bin

ECE = 0.0009 (near-perfect aggregate calibration). The elevated MCE reflects softmax overconfidence in the high-probability bin, driven by the dominant Benign class (50.8% of flows). The decision layer partially mitigates this by requiring hierarchy consistency before committing to a fine label. Thresholds τ_c and τ_f are fixed post-calibration; periodic recalibration or temperature scaling [6] under distribution shift is future work.

VII. Discussion

A. The Cost-Capability Tradeoff

For CNN1D, the hierarchy costs 1.04 F1 points in exchange for three capabilities flat classifiers structurally cannot provide: 99.97% selective fine accuracy at 94.5% coverage; 98–99% coarse accuracy on exactly the samples most likely to produce silent flat errors; and 62.1% novel alarm rate without any explicit OOD training. For AttentionDNN the cost is neutral (+0.23 pp mean). A natural missing baseline is HiE-style soft reweighting [12] ($\tilde{p}_f(k) \propto p_f(k) \cdot p_c(\phi(k))$), which would mitigate coarse head error recovery at the cost of the hard guarantee; comparison is future work. A natural missing baseline is HiE-style soft reweighting [12], which replaces Stage 2 hard masking with multiplicative down-weighting: $\tilde{p}_f(k) \propto p_f(k) \cdot p_c(\phi(k))$. This variant would reduce recovery failures under coarse head errors at the cost of the hard consistency guarantee; an empirical comparison of hard vs. soft masking is immediate future work.

B. Flat & Abstention Comparison

To isolate the hierarchical contribution from selective prediction, we trained an independent flat CNN1D (identical backbone, no coarse head or consistency loss) and equipped it with max-softmax thresholding at the HiNeS-IDS operating point. Table XI reports the comparison at CNN1D seed 42.

TABLE XI
SELECTIVE PREDICTION COMPARISON (CNN1D, SEED 42). †: FLAT SOFTMAX TOO CONCENTRATED TO REACH 94.5% FINE COVERAGE.

Method	Fine cov.	Sel. acc.	Coarse tier
HiNeS-IDS	94.5%	99.98%	4.0% (98.6%)
Flat+MaxSoftmax	60.9%†	99.998%	—
Flat (no abstain)	100%	97.3% F1	—

Both systems achieve near-identical selective fine accuracy (≈99.98%). The flat softmax is too concentrated to reach 94.5% coverage, jumping from ∼100% to 60.9% with no intermediate—confirming that the hierarchy's unique contribution is the *coarse-only tier*: converting silent abstentions into 98.6%-accurate family-level alerts.

C. On the Missing HiE Baseline

A natural absent comparison is HiE-style soft reweighting [12], which replaces Stage 2 hard masking with multiplicative down-weighting: $\tilde{p}_f(k) \propto p_f(k) \cdot p_c(\phi(k))$. This variant reduces recovery failures under coarse head errors at the cost of the hard consistency guarantee; it requires no retraining and can be evaluated as a post-hoc inference modification. An empirical hard-vs-soft masking comparison is immediate future work.

D. Capacity Interference in MTL

The capacity-dependent λ_{con} interaction is the paper's most theoretically informative result. CNN1D (265K params) benefits up to λ_{con}=0.3, where the consistency loss acts as a regulariser forcing shared structure into the backbone. AttentionDNN (187K params) degrades beyond λ_{con}=0.01, where the secondary objective consumes gradient bandwidth needed for the primary task. This illustrates MTL interference [26]: for models with $\gtrsim$ 250K parameters λ_{con}=0.3 is beneficial; for smaller models, λ_{con}=0.

E. Emergent Open-Set Geometry

The 60-point gap between centroid (69.5%) and softmax (9.1%) family assignment on novel UDP Flood is not engineered—it emerges because the consistency loss implicitly trains the backbone to organise embeddings by attack family. Novel subtypes with family-level features cluster near the correct centroid even without training exposure. This is qualitatively distinct from post-hoc energy or Mahalanobis scoring applied to a representation never trained to be open-set aware.

F. Temporal Leakage as a Community Concern

The 64-point F1 drop under temporal evaluation is a finding about CICIoT2023, not about HiNeS-IDS. The non-monotonic per-block pattern (B2 lowest at 0.211) rules out gradual drift and points to burst-like attack sessions that leak across random splits. Random-split results on this dataset should be treated as upper bounds, not deployment estimates. We report both protocols to document the gap rather than silently adopting the inflated convention [27].

G. Limitations

Temporal adaptation (primary): static thresholds drift under distribution shift; continual learning or sliding-window recalibration is needed. **Hard masking**: coarse head errors suppress correct fine classes; soft-reweighting (HiE-style) is the natural mitigation. **Scope**: single-dataset, two-seed CNN1D; cross-dataset and leave-one-class-out OOD evaluation are immediate future work. **Taxonomy sensitivity**: the five keyword rules are dataset-specific and require manual adaptation for new label schemas. Robustness to noisy or ambiguous coarse assignments has not been evaluated; ontology-driven or learned mappings from richer hierarchies (e.g., MITRE ATT&CK sub-techniques) are a natural generalisation. **Taxonomy mapping**: the keyword rules are CICIoT2023-specific; a mislabelled coarse assignment propagates into Stage 2 masking errors. Ontology-driven or learned mappings are a natural generalisation. **Taxonomy sensitivity**: the keyword mapping is CICIoT2023-specific and requires manual adaptation for new datasets; robustness to label noise or ambiguous class names is unvalidated. Ontology-driven or learned mappings from richer hierarchies (e.g., MITRE ATT&CK sub-techniques) are a natural generalisation. **Taxonomy sensitivity:** the five keyword rules are CICIoT2023-specific and require manual adaptation for new datasets. Robustness to noisy or ambiguous labels is unevaluated; ontology-driven or learned mappings (e.g., MITRE ATT&CK sub-techniques) are a natural generalisation.

VIII. Conclusion

We presented HiNeS-IDS, a dual-head neural NIDS with a three-stage taxonomy-constrained decision layer that guarantees zero hierarchy violations by construction. The consistency loss improves CNN1D F1 by $+1.05$ pp while exposing a capacity-dependent MTL interaction that degrades AttentionDNN beyond $\lambda_{\text{con}}{=}0.01$. The coarse-only tier achieves 98–99% family accuracy on samples where flat classifiers produce 65–71% fine accuracy. Hierarchical multi-task training produces an emergent embedding geometry that assigns 69.5% of novel samples to the correct family—60 points above the softmax head. A strict temporal evaluation reveals a 64-point F1 collapse, identifying random-split optimism as a pervasive bias in IoT NIDS benchmarking. The flat+abstention comparison confirms that selective accuracy is table-stakes achievable by thresholding; the coarse-only tier is the unique, operationally irreplaceable contribution of the hierarchical decision layer. HiNeS-IDS does not merely predict more carefully—it predicts at the right level of abstraction for each input. A natural missing baseline is HiE-style soft reweighting [12], which replaces Stage 2 hard masking with $\tilde{p}_f(k) \propto p_f(k) \cdot p_c(\phi(k))$. This would reduce recovery failures under coarse-head errors at the cost of the hard consistency guarantee; an empirical comparison is immediate future work.

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